

Nikolai Slavov

CONTACT INFORMATION

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ORCID ID: [0000-0003-2035-1820](https://orcid.org/0000-0003-2035-1820)

Academic Laboratory: slavovlab.net
Parallel Squared Institute: parallelsq.org
SCP Center: center.single-cell.net

EDUCATION

Princeton University, Princeton, USA
Ph.D. in Molecular and Quantitative Cell Biology **06.2010**
M.A. in Molecular Biology **02.2006**
Mentor: David Botstein
Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations
Massachusetts Institute of Technology (MIT), Cambridge, USA
B.S. in Biology **06.2004**

RESEARCH & ACADEMIC APPOINTMENTS

Parallel Squared Technology Institute (PTI), MA, USA
Founding Director **2023–present**
Northeastern University, Boston, USA
Professor, Department of Bioengineering **2024–present**
Faculty Fellow, Barnett Institute **2017–present**
Affiliated Faculty, Department of Chemistry and Chemical Biology **2016–present**
Affiliated Faculty, Department of Biology **2016–present**
Associate Professor, Department of Bioengineering **2021–2024**
Assistant Professor, Department of Bioengineering **2015–2021**
Harvard University, Cambridge, USA
Research Fellow, Departments of Systems Biology and Statistics **2013–2015**
Massachusetts Institute of Technology (MIT), Cambridge, USA
Research Fellow, van Oudenaarden laboratory, Departments of Biology and Physics **2011–2013**

AWARDS & FELLOWSHIPS

✓ [HUPO Discovery in Proteomic Sciences Award](#) **2022**
✓ College of Engineering Faculty Fellow **2022**
✓ [Allen Distinguished Investigator](#) (\$ 1, 500, 000) **2020**
✓ Excellence in Mentoring Award **2018**
✓ [NIH Director's New Innovator Award](#) (\$ 2, 352, 750) **2016**
✓ Broad Institute of Harvard and MIT SPARC Award **2014**
✓ Princeton University Dean's Award **2010**
✓ IRCSET Postgraduate Research Fellowship (72, 000 €) **2007**
✓ Finalist in the Young European Entrepreneur Competition **2006**

- ✓ Princeton Graduate Fellowship (\$ 45, 000) **2004**
- ✓ MIT Undergraduate Fellowship (\$ 125, 000) **2001**
- ✓ Eureka Fellowship for Academic Excellence (10, 000 €) **2000**
- ✓ Bronze Medal in the 31st International Chemistry Olympiad (*IChO*) **1999**
- ✓ National Diploma for Exceptional Achievements in Chemistry **1999**

SELECTED
PEER REVIEWED
PUBLICATIONS

[OA] – Open Access article

✉ – Corresponding author

[Google Scholar Profile](#)

2025

2025

- ✓ **Slavov N**✉
Single-Cell Proteomic Technologies: Tools in the quest for principles
Annual Review of Biophysics(in press), DOI: [10.48550/arXiv.2506.18198](https://doi.org/10.48550/arXiv.2506.18198)
- ✓ **Slavov N**✉ *et al.*
Artificial intelligence in metabolic research
Nat Metab. 2025, DOI: [10.1038/s42255-025-01395-7](https://doi.org/10.1038/s42255-025-01395-7)
- ✓ Liu Y, ..., **Slavov N**, Murn J
Composition and RNA binding specificity of metazoan RNase MRP
Nucleic Acids Res 2025, 27;53(16):gkaf829 DOI: [10.1093/nar/gkaf829](https://doi.org/10.1093/nar/gkaf829)
- ✓ **Slavov N**✉
Unlocking the Potential of Single-Cell Omics
J. Proteome Res. 2025, 24, 1481 – 1481 DOI: [0.1021/acs.jproteome.5c00197](https://doi.org/0.1021/acs.jproteome.5c00197)
- ✓ Leduc A, Xu Y, Shipkovenska G, Dou Z, **Slavov N**✉
Limiting the impact of protein leakage in single-cell proteomics
Nature Communications, 5;16(1):4169 DOI: [10.1038/s41467-025-56736-7](https://doi.org/10.1038/s41467-025-56736-7)
- ✓ Rengarajan S, Derks J, Bellott D, **Slavov N**, Page DC
Post-transcriptional cross- and auto-regulation buffer expression of the human RNA helicases DDX3X and DDX3Y *Genome Research*, DOI: [10.1101/gr.279707.124](https://doi.org/10.1101/gr.279707.124)

2024

2024

- ✓ Leduc A., Koury L., Cantlon J., Khan S., **Slavov N**✉
Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP
Nature Protocols, DOI: [10.1038/s41596-024-01033-8](https://doi.org/10.1038/s41596-024-01033-8)
- ✓ SenNet Consortium, **Slavov N**, *et al.*
SenNet recommendations for detecting senescent cells in different tissues
Nat Rev Mol Cell Biol 25(12):1001-1023, DOI: [10.1038/s41580-024-00738-8](https://doi.org/10.1038/s41580-024-00738-8)

- ✓ Galatidou S, Petelski A, Pujol A, Lattes K,..., **Slavov N**, Barragan M
Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes
Molecular Human Reproduction, DOI: [10.1093/molehr/gaae023](https://doi.org/10.1093/molehr/gaae023) [OA]
- ✓ Rogers ZJ, Colombani T, Khan S,..., **Slavov N**, Bencherif SA
Controlling Pericellular Oxygen Tension in Cell Culture Reveals Distinct Breast Cancer Responses to Low Oxygen Tensions
Adv Sci. e2402557, DOI: [10.1002/advs.202402557](https://doi.org/10.1002/advs.202402557) [OA]
- ✓ Sipe SS, **Slavov N**✉
Single-Cell Proteomics Accelerates toward Proteoforms
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.4c00290](https://doi.org/10.1021/acs.jproteome.4c00290)
- ✓ Khan S, Conover R, Asthagiri AR, **Slavov N**✉
Dynamics of single-cell protein covariation during epithelial-mesenchymal transition
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.4c00277](https://doi.org/10.1021/acs.jproteome.4c00277) [OA]

2023

2023

- ✓ Wallmann G., Leduc A., **Slavov N**✉
Data-Driven Optimization of DIA Mass Spectrometry by DO-MS
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.3c00177](https://doi.org/10.1021/acs.jproteome.3c00177) [OA]
- ✓ **Slavov N**✉
Single-cell proteomics: quantifying post-transcriptional regulation during development with mass-spectrometry
Development, 150 (13): dev201492, DOI: [10.1242/dev.201492](https://doi.org/10.1242/dev.201492)
- ✓ van den Berg PR, Bérenger-Currias NMLP, Budnik B, **Slavov N**✉, Semrau S
Integration of a multi-omics stem cell differentiation dataset using a dynamical model
PLOS Genetics **19** (5) :1-23 DOI: [10.1371/journal.pgen.1010744](https://doi.org/10.1371/journal.pgen.1010744) [OA]
- ✓ Huffman RG, Leduc A, Wichmann C, di Gioia M, Borriello F, Specht H, Derks J, Khan S,..., **Slavov N**✉
Prioritized mass spectrometry increases the depth, sensitivity and data completeness of single-cell proteomics
Nature Methods, **20**, 714-722 DOI: [10.1038/s41592-023-01830-1](https://doi.org/10.1038/s41592-023-01830-1) [OA]
- ✓ Gatto L, Aebersold R, Cox J, Demichev V, Derks J, Emmott E, Franks AM, Ivanov AR,..., **Slavov N**✉
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
Nature Methods **20**, 375–386 DOI: [10.1038/s41592-023-01785-3](https://doi.org/10.1038/s41592-023-01785-3) [OA]
- ✓ MacCoss MJ, Alfaro J, Faivre, D, Wu C, Wanunu M, **Slavov N**✉
Sampling proteomes by emerging single-molecule and mass-spectrometry methods
Nature Methods **20**, 339–346, DOI: [10.1038/s41592-023-01802-5](https://doi.org/10.1038/s41592-023-01802-5)
- ✓ **Slavov N**✉
Great Gains in Mass Spectrometry Data Interpretation
Journal of Proteome Research, **22** 3, 659 DOI: [10.1021/acs.jproteome.3c00099](https://doi.org/10.1021/acs.jproteome.3c00099)

- ✓ Derks J, **Slavov N**[✉]
Strategies for increasing the depth and throughput of protein analysis by plexDIA
Journal of Proteome Research **22** 697-705, DOI: [10.1021/acs.jproteome.2c00721](https://doi.org/10.1021/acs.jproteome.2c00721) [OA]

2022

2022

- ✓ Leduc A, Huffman RG, Cantlon J, Khan S, **Slavov N**[✉]
Exploring functional protein covariation across single cells using nPOP
Genome Biology, **23**, 261 DOI: [10.1186/s13059-022-02817-5](https://doi.org/10.1186/s13059-022-02817-5) [OA]
↪ [Single-Cell Proteomics Bypasses Bottlenecks, Sways Skeptics](#), highlight by *GEN*
- ✓ Petelski AA, **Slavov N**, Specht H
Single-Cell Proteomics Preparation for Mass Spectrometry Analysis Using Freeze-Heat Lysis and an Isobaric Carrier
J. Vis. Exp., (190), DOI: [10.3791/63802](https://doi.org/10.3791/63802)
- ✓ Derks J, Leduc A, Wallmann G, Huffman G, Willetts M, Khan S, Specht H,..., **Slavov N**[✉]
Increasing the throughput of sensitive proteomics by plexDIA
Nature Biotechnology, DOI: [10.1038/s41587-022-01389-w](https://doi.org/10.1038/s41587-022-01389-w)
↪ [Sensitive protein analysis with plexDIA](#), highlight by *Nature Methods*
↪ [Uncovering Proteomic Patterns One Cell at a Time](#), highlight by *GEN*
- ✓ Specht H, **Slavov N**[✉]
Beyond Protein Sequence: Protein Isomerization in Alzheimer's Disease
Journal of Proteome Research **21** (2) :299-300, DOI: [10.1021/acs.jproteome.2c00016](https://doi.org/10.1021/acs.jproteome.2c00016)
- ✓ SenNet Consortium, **Slavov N**
NIH SenNet Consortium to map senescent cells throughout the human lifespan to understand physiological health
Nat aging 1090–1100 DOI: [10.1038/s43587-022-00326-5](https://doi.org/10.1038/s43587-022-00326-5)
- ✓ **Slavov N**[✉]
Counting protein molecules for single-cell proteomics
Cell **185** (2) :232-234, DOI: [10.1016/j.cell.2021.12.013](https://doi.org/10.1016/j.cell.2021.12.013)
- ✓ **Slavov N**[✉]
Learning from natural variation across the proteomes of single cells
PLoS Biology **20**(1): e3001512, DOI: [10.1371/journal.pbio.3001512](https://doi.org/10.1371/journal.pbio.3001512) [OA]
- ✓ **Slavov N**[✉]
Scaling up single-cell proteomics
Molecular & Cellular Proteomics **21**(1), 100179, DOI: [10.1016/j.mcpro.2021.100179](https://doi.org/10.1016/j.mcpro.2021.100179) [OA]

2021

2021

- ✓ Khoury L, **Slavov N**[✉]
Comprehensive Identification of Regulatory Protein Networks
Journal of Proteome Research **20**(11): 4913-4914, DOI: [10.1021/acs.jproteome.1c00813](https://doi.org/10.1021/acs.jproteome.1c00813)

- ✓ He L, *et al.*, **Slavov N**, Marneros A,
Global characterization of macrophage polarization mechanisms and identification of M2-type polarization inhibitors
Cell Reports 37(5):109955, DOI: [10.1016/j.celrep.2021.109955](https://doi.org/10.1016/j.celrep.2021.109955)
- ✓ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**
Multiplexed single-cell proteomics using SCoPE2
Nature Protocols 16, 5398–5425, DOI: [10.1038/s41596-021-00616-z](https://doi.org/10.1038/s41596-021-00616-z)
- ✓ **Slavov N**
Driving Single Cell Proteomics Forward with Innovation
Journal of Proteome Research 20, 11, 4915–4918, DOI: [10.1021/acs.jproteome.1c00639](https://doi.org/10.1021/acs.jproteome.1c00639)
- ✓ **Slavov N**
Measuring Protein Shapes in Living Cells
Journal of Proteome Research DOI: [10.1021/acs.jproteome.1c00376](https://doi.org/10.1021/acs.jproteome.1c00376)
- ✓ **Slavov N**
Increasing proteomics throughput
Nature Biotechnology, 39, 809–810, DOI: [10.1038/s41587-021-00881-z](https://doi.org/10.1038/s41587-021-00881-z)
- ✓ **Slavov N** *et al.*
Voices of biotech research: Single-cell proteomics
Nature Biotechnology, 39, 281–286, DOI: [10.1038/s41587-021-00847-1](https://doi.org/10.1038/s41587-021-00847-1)
- ✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, Serra M, Kharchenko P, Koller A & **Slavov N**
Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
Genome Biology, 22, 50 DOI: [10.1186/s13059-021-02267-5](https://doi.org/10.1186/s13059-021-02267-5) [OA]
↪ [Towards resolving proteomes in single cells](#), highlight by *Nature Methods*

2020

2020

- ✓ Specht H, **Slavov N**
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.0c00675](https://doi.org/10.1021/acs.jproteome.0c00675)
- ✓ Petelski AA **Slavov N**
Analyzing ribosome remodeling in health and disease
Proteomics, 20, (17) :2000039 DOI: [10.1002/pmic.202000039](https://doi.org/10.1002/pmic.202000039)
- ✓ **Slavov N**
Single-cell protein analysis by mass-spectrometry
Current Opinion in Chemical Biology, 60, 1 - 9 DOI: [10.1016/j.cbpa.2020.04.018](https://doi.org/10.1016/j.cbpa.2020.04.018)
- ✓ **Slavov N**
Unpicking the proteome in single cells
Science, 367 (6477) :512 - 513, DOI: [10.1126/science.aaz6695](https://doi.org/10.1126/science.aaz6695)

2019

2019

- ✓ Slavov N[✉] *et al.*
Voices in methods development: Single-cell proteomics
Nature Methods, 16, 945 - 951, DOI: [10.1038/s41592-019-0585-6](https://doi.org/10.1038/s41592-019-0585-6)
- ✓ Huffman RG, Chen AT, Specht H & Slavov N[✉]
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
Journal of Proteome Research, 18, 6, 2493 - 2500, DOI: [10.1021/acs.jproteome.9b00039](https://doi.org/10.1021/acs.jproteome.9b00039)
⇒ [Technology feature](#) highlight by *Nature Methods*
- ✓ Chen AT, Franks A & Slavov N[✉]
DART-ID increases single-cell proteome coverage
PLoS Computational Biology, 5(7):e1007082 DOI: [10.1371/journal.pcbi.1007082](https://doi.org/10.1371/journal.pcbi.1007082) [OA]
⇒ [The single cell proteomics revolution](#), editorial highlight by *Bioanalysis Zone*
- ✓ Emmott EP, Jovanovic M & Slavov N[✉]
Approaches for studying ribosome specialization
Trends in Biochemical Sciences, 44(5):478 - 479, DOI: [10.1016/j.tibs.2019.01.008](https://doi.org/10.1016/j.tibs.2019.01.008)
- ✓ Emmott EP, Jovanovic M & Slavov N[✉]
Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, 44(2):95 - 109, DOI: [10.1016/j.tibs.2018.10.009](https://doi.org/10.1016/j.tibs.2018.10.009)
- ✓ Malioutov D., Chen T., Jaffe J., Airoidi E., Budnik B. & Slavov N[✉]
Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, 18(1):162 - 168, DOI: [10.1074/mcp.TIR118.000947](https://doi.org/10.1074/mcp.TIR118.000947)

2018

2018

- ✓ Specht H. & Slavov N[✉]
Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: [10.1021/acs.jproteome.8b00257](https://doi.org/10.1021/acs.jproteome.8b00257)
⇒ [Innovations in Proteomics: The Drive to Single Cells](#), editorial highlight
⇒ [Through the Looking Glass of Single Cell Proteomics](#), highlight by *Technology Networks*
- ✓ Budnik B., Levy E., Harmange G. & Slavov N[✉]
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: [10.1186/s13059-018-1547-5](https://doi.org/10.1186/s13059-018-1547-5) [OA]
⇒ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
⇒ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
⇒ [Single-cell proteomics](#) highlight by *the NIH Director's office*
⇒ [Technology feature](#) highlight by *Nature Methods*
⇒ [The single cell proteomics revolution](#), editorial highlight by *Bioanalysis Zone*
- ✓ Levy E. & Slavov N[✉]
Single cell protein analysis for systems biology
Essays in Biochemistry, 62(4):595 - 605, DOI: [10.1042/EBC20180014](https://doi.org/10.1042/EBC20180014)

2017

2017

- ✓ Franks A., Airoidi E.M., Slavov N[✉]
Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: [10.1371/journal.pcbi.1005535](https://doi.org/10.1371/journal.pcbi.1005535) [OA]
⇒ [How Statistics Weakened mRNA's Predictive Power](#) highlight by *The Scientist*

- ✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N**, Pasparakis M, Balachandran S, Kelliher M, Mecsas J, Degterev A. Kinase Activities of RIPK1 and RIPK3 Can Direct IFN- β Synthesis Induced by Lipopolysaccharide. *The Journal of Immunology*, 198(11):4435-4447, DOI: [10.4049/jimmunol.1601717](https://doi.org/10.4049/jimmunol.1601717)

2016

2016

- ✓ Klionsky, *et al.*, **Slavov N**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, 12(1):1-222, DOI: [10.1080/15548627.2015.1100356](https://doi.org/10.1080/15548627.2015.1100356)
- ✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3):e0150605, DOI: [10.1371/journal.pone.0150605](https://doi.org/10.1371/journal.pone.0150605) [OA]

2015

2015

- ✓ **Slavov N** 
Making the most of peer review
eLife, 4:e12708, DOI: [10.7554/eLife.12708](https://doi.org/10.7554/eLife.12708) [OA]
- ✓ **Slavov N** , Semrau S., Airolidi E.M., Budnik B., van Oudenaarden A.
Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: [10.1016/j.celrep.2015.09.056](https://doi.org/10.1016/j.celrep.2015.09.056) [OA]
→ [All Ribosomes Are Created Equal. Really?](#), highlight by *Trends in Biochemical Sciences*
→ [New direction for tissue engineering and cancer therapeutics](#), highlight by *Science X*
- ✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N**, *et al.*
De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: [10.1016/j.cmet.2015.04.003](https://doi.org/10.1016/j.cmet.2015.04.003)

Started Slavov Lab at Northeastern University, August 2015

2014

2014

- ✓ Malioutov D. , **Slavov N** 
Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 [OA]
- ✓ **Slavov N** , Budnik B., Schwab D., Airolidi E., van Oudenaarden A. 
Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 [OA]

2013

2013

- ✓ Slavov N[✉], Carey, J., Linse, S.[✉]
Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 [OA]
- ✓ Slavov N[✉], Botstein D.[✉]
Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Deregulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 [OA]

2012

2012

- ✓ Slavov N[✉], van Oudenaarden A.[✉]
How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 [OA]
- ✓ Slavov N[✉], Airoidi E.M., van Oudenaarden A., Botstein D.[✉]
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 [OA]

2011

2011

- ✓ Slavov N[✉], Macinskas J., Caudy A., Botstein D.[✉]
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 [OA]
- ✓ Slavov N[✉], Botstein D.[✉]
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 [OA]

2010

2010

- ✓ Slavov N[✉]
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 [OA]
- ✓ Slavov N[✉], Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University [OA]
- ✓ Silverman SJ., Slavov N, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.[✉]
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 [OA]

2009 and before

2009 and before

- ✓ Slavov N[✉], Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079-4084 (“In This Issue” article) [\[OA\]](#)
- ✓ Tagkopoulos I. [✉], Slavov N[✉], Kung S. [✉] (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5th IEEE Symposium on Bioinformatics and Bioengineering (BIBE’05).

PATENTS

- Slavov N[✉], *et al.*, Miniaturized Proteomic Sample Preparation. U.S. Patent Application No. 18/556,655.
- Slavov N[✉], *et al.*, Methods of multiplexed data-independent acquisition for proteomics. U.S. Patent Application No. 17/806,468.
- Slavov N[✉], *et al.*, Mass spectrometry technique for single cell proteomics, US Patent No. 11,719,703 issued on 8/8/2023.

PUBLISHED

PREPRINTS

GOOGLE SCHOLAR

- ▷ Leduc A, Shipkovenska G, Xu Y, Franks A, Slavov N[✉] (2025)
The regulation of protein abundance across single cells in a mammalian tissue
bioRxiv DOI: [10.1101/2025.09.17.676955](https://doi.org/10.1101/2025.09.17.676955)
- ▷ Slavov N[✉] (2025)
Single-Cell Proteomic Technologies: Tools in the quest for principles
arXiv DOI: [10.48550/arXiv.2506.18198](https://doi.org/10.48550/arXiv.2506.18198)
- ▷ Derks, J., McDonnell, K., Wamsley, N., Stewart, P., Yeh, M., Specht, H., Slavov N[✉] (2025)
Increasing mass spectrometry throughput using time-encoded sample multiplexing
bioRxiv DOI: [10.1101/2025.05.22.655515](https://doi.org/10.1101/2025.05.22.655515)
- ▷ McDonnell, K., Wamsley, N., Derks, J., Sipe, S., Yeh, M., Specht, H., Slavov N[✉] (2025)
JMod: Joint modeling of mass spectra for empowering multiplexed DIA proteomics
bioRxiv DOI: [10.1101/2025.05.22.655512](https://doi.org/10.1101/2025.05.22.655512)
- ▷ Specht, H., Yeh, M., Sipe, S., Barnes-Seeman, D., Adamo, M., McDonnell, K., Agius, M. P., Friedrich, C., Pang, W. K., Huang, Y., Shiva Raju, K., Vuong, W., Lee, M. A., Yesilcimen, A., Pentelute, B. L., Slavov N[✉] (2025)
PSMtags improve peptide sequencing and throughput in sensitive proteomics
bioRxiv, DOI: [10.1101/2025.05.22.655509](https://doi.org/10.1101/2025.05.22.655509)

- ▷ Leduc A, **Slavov N**[✉] (2025)
Protein degradation and growth dependent dilution substantially shape mammalian proteomes
bioRxiv DOI: [10.1101/2025.02.10.637566](https://doi.org/10.1101/2025.02.10.637566)
- ▷ Khan S, Elcheikhali M, Leduc A, Huffman R, Derks J, Franks A, **Slavov N**[✉] (2024)
Inferring post-transcriptional regulation within and across cell types in human testis
bioRxiv DOI: [10.1101/2024.10.08.617313](https://doi.org/10.1101/2024.10.08.617313)
- ▷ Iwamoto-Stohl LK, Petelski AA, ... **Slavov N**, Zernicka-Goetz (2024)
Proteome asymmetry in mouse and human embryos before fate specification
bioRxiv DOI: [10.1101/2024.08.26.609777](https://doi.org/10.1101/2024.08.26.609777)
- ▷ Tsour S, Machne R, Leduc A, Widmer S, **Slavov N**[✉] (2024)
Alternate RNA decoding results in stable and abundant proteins in mammals
bioRxiv DOI: [10.1101/2024.08.26.609665](https://doi.org/10.1101/2024.08.26.609665)
- ▷ Leduc A, Xu Y, Shipkovenska G, Dou Z, **Slavov N**[✉] (2024)
Limiting the impact of protein leakage in single-cell proteomics
bioRxiv DOI: [10.1101/2024.07.26.605378](https://doi.org/10.1101/2024.07.26.605378)
- ▷ Derks J, Jonson T, Leduc A, Khan S, Khoury L, Rafiee M, **Slavov N**[✉] (2024) Single-nucleus proteomics identifies regulators of protein transport
bioRxiv DOI: [10.1101/2024.06.17.599449](https://doi.org/10.1101/2024.06.17.599449)
- ▷ Galatidou S, Petelski A, Pujol A, ..., **Slavov N**[✉], Barragan M (2024)
Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes
bioRxiv DOI: [10.1101/2024.05.23.595547](https://doi.org/10.1101/2024.05.23.595547)
- ▷ Khan S, Conover R, Asthagiri AR, **Slavov N**[✉] (2023)
Dynamics of single-cell protein covariation during epithelial-mesenchymal transition
bioRxiv, DOI: [10.1101/2023.12.21.572913](https://doi.org/10.1101/2023.12.21.572913) [OA]
- ▷ Leduc A., Koury L., Cantlon J., **Slavov N**[✉] (2023)
Massively parallel sample preparation for multiplexed single-cell proteomics using nPOP
bioRxiv, DOI: [10.1101/2023.11.27.568927](https://doi.org/10.1101/2023.11.27.568927) [OA]
- ▷ Leduc A, Harens H, **Slavov N**[✉] (2023)
Modeling and interpretation of single-cell proteogenomic data
arXiv DOI: [10.48550/arXiv.2308.07465](https://doi.org/10.48550/arXiv.2308.07465)
- ▷ Wallmann G., Leduc A., **Slavov N**[✉] (2023)
Data-Driven Optimization of DIA Mass-Spectrometry by DO-MS
bioRxiv, DOI: [10.1101/2023.02.02.526809](https://doi.org/10.1101/2023.02.02.526809) [OA]

- ▷ Derks J, **Slavov N** (2022)
Strategies for increasing the depth and throughput of protein analysis by plexDIA
bioRxiv, DOI: [10.1101/2022.11.05.515287](https://doi.org/10.1101/2022.11.05.515287) [OA]
- ▷ Gatto L, ..., **Slavov N** (2022)
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
arXiv, DOI: [10.48550/arXiv.2207.10815](https://doi.org/10.48550/arXiv.2207.10815) [OA]
- ▷ MacCoss MJ, Alfaro J, Wanunu M, Faivre DA, **Slavov N** (2022)
Sampling proteomes by emerging single-molecule and mass-spectrometry methods
arXiv, DOI: [10.48550/arXiv.2208.00530](https://doi.org/10.48550/arXiv.2208.00530) [OA]
- ▷ Huffman RG, Leduc A, ..., **Slavov N** (2022)
Prioritized single-cell proteomics reveals molecular and functional polarization across primary macrophages
bioRxiv, DOI: [10.1101/2022.03.16.484655](https://doi.org/10.1101/2022.03.16.484655) [OA]
- ▷ Derks J, Leduc A, Huffman RG, Specht H, Ralser M, Demichev V, **Slavov N** (2021)
Increasing the throughput of sensitive proteomics by plexDIA
bioRxiv, DOI: [10.1101/2021.11.03.467007](https://doi.org/10.1101/2021.11.03.467007) [OA]
- ▷ Leduc A, Huffman RG, **Slavov N** (2021)
Droplet sample preparation for single-cell proteomics applied to the cell cycle
bioRxiv, DOI: [10.1101/2021.04.24.441211](https://doi.org/10.1101/2021.04.24.441211) [OA]
- ▷ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N** (2021)
Multiplexed single-cell proteomics using SCoPE2
bioRxiv, DOI: [10.1101/2021.03.12.435034](https://doi.org/10.1101/2021.03.12.435034) [OA]
- ▷ Specht H, **Slavov N** (2020)
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
bioRxiv, DOI: [10.1101/2020.08.24.264994](https://doi.org/10.1101/2020.08.24.264994) [OA]
- ▷ Petelski AA, **Slavov N** (2020)
Analyzing ribosome remodeling in health and disease
arXiv, [arXiv:2007.05839](https://arxiv.org/abs/2007.05839) [OA]
- ▷ **Slavov N** (2020)
Single-cell protein analysis by mass-spectrometry
arXiv, [arXiv:2004.02069](https://arxiv.org/abs/2004.02069) [OA]
- ▷ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, *et al.* & **Slavov N** (2019)
Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity *bioRxiv*, DOI: [10.1101/665307](https://doi.org/10.1101/665307) [OA]
↪ [Over 1,000 SINGLE CELL PROTEOMES!](#), highlight by *Proteomics News*
↪ [Technology feature](#) highlight by *Nature Methods*

- ▷ Specht H, Emmott E, Koller A, **Slavov N**[✉] (2019)
High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
bioRxiv, DOI: [10.1101/665307](https://doi.org/10.1101/665307) [OA]
- ▷ Huffman G, Specht H, Chen AT, **Slavov N**[✉] (2019)
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
bioRxiv, DOI: [10.1101/512152](https://doi.org/10.1101/512152) [OA]
- ▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N**[✉] (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: [10.1101/399774](https://doi.org/10.1101/399774) [OA]
- ▷ Chen A, Franks A, **Slavov N**[✉] (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: [10.1101/399121](https://doi.org/10.1101/399121) [OA]
- ▷ Levy E. & **Slavov N**[✉] (2018)
Single cell protein analysis for systems biology
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26965v1](https://doi.org/10.7287/peerj.preprints.26965v1) [OA]
- ▷ Emmott EP, Jovanovic M & **Slavov N**[✉] (2018)
Ribosome stoichiometry: from form to function
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26991v1](https://doi.org/10.7287/peerj.preprints.26991v1) [OA]
- ▷ Specht H. & **Slavov N**[✉] (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
PeerJ Preprint, DOI: [10.7287/peerj.preprints.26821v1](https://doi.org/10.7287/peerj.preprints.26821v1) [OA]
- ▷ Malioutov D., Chen T., Jaffe J., Airoidi E., Carr S., Budnik B., **Slavov N**[✉] (2017)
Quantifying homologous proteins and proteoforms
bioRxiv, DOI: [10.1101/168765](https://doi.org/10.1101/168765) [OA]
- ▷ Berg P., Budnik B., **Slavov N**[✉], Semrau S. (2017)
Dynamic post-transcriptional regulation during embryonic stem cell differentiation
bioRxiv, DOI: [10.1101/123497](https://doi.org/10.1101/123497) [OA]
- ▷ Budnik B., Levy E., **Slavov N**[✉] (2017)
Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
bioRxiv, DOI: [10.1101/102681](https://doi.org/10.1101/102681) [OA]
 - ↪ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
 - ↪ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
 - ↪ [Interview](#), highlight by *Front Line Genomics*

- ▷ **Slavov N**[✉] (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: [10.7287/peerj.preprints.1504v1](https://doi.org/10.7287/peerj.preprints.1504v1) [OA]
- ▷ Franks A., Airoidi E.M., **Slavov N**[✉] (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: [10.1101/020206](https://doi.org/10.1101/020206) [OA]
- ▷ **Slavov N**[✉], Botstein, D., Caudy A. (2014)
Extensive regulation of metabolism and growth during the cell division cycle
bioRxiv, DOI: [10.1101/005629](https://doi.org/10.1101/005629) [OA]
- ▷ **Slavov N**[✉], Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A. (2014)
Differential stoichiometry among core ribosomal proteins
bioRxiv, DOI: [10.1101/005553](https://doi.org/10.1101/005553) [OA]

OPINIONS &
POPULAR SCIENCE
PUBLICATIONS

- ▷ **Slavov N**[✉] (2018) We Must Demand Evidence of Peer Review
The Scientist, [Link](#)
- ▷ **Slavov N**[✉] (2017) What was rate limiting was the idea, not the technology
Front Line Genomics, [Link](#)
- ▷ **Slavov N**[✉] (2015) Making the most of peer review
eLife, [Link](#)
- ▷ **Slavov N**[✉] (2014) Accomplishments Over Accolades!
The Scientist, [Link](#)
- ▷ **Slavov N**[✉] (2014) Discoveries lie hidden behind the façade of popular assumptions
Cell Reporter, [Link](#)
- ▷ **Slavov N**[✉] (2013) The Best Projects Are Least Obvious
The Scientist, [Link](#)
- ▷ **Slavov N**[✉] (2013) A Wonderful Christmas with Mysterious Oscillations
PubChase, [Link](#)
- ▷ **Slavov N**[✉] (2012) The Mission of MIT
The MIT Tech, [Link](#)